JEE MAIN / PREVIOUS YEAR PAPER

JEE Main 2025 23rd January Evening Shift

Question Paper with Answer Key

75	ENGLISH	DARK
QUESTIONS	LANGUAGE	FORMAT

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Mathematics	25 questions
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Answer key	Included at the end

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QR codes open the DekhoExams website.

Chemistry 25 questions

Q1. MCQ (Single Correct Answer)

pH of water is 7 at $25^{\circ}\mathrm{C}$. If water is heated to $80^{\circ}\mathrm{C}$., it's pH will :

(A) Decrease

(B) Remains the same

(C) Increase

(D) H^{+}concentration increases, OH^{-}concentration decreases

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Q2. MCQ (Single Correct Answer)

When a non-volatile solute is added to the solvent, the vapour pressure of the solvent decreases by 10 mm of Hg . The mole fraction of the solute in the solution is 0.2 . What would be the mole fraction of the solvent if decrease in vapour pressure is 20 mm of Hg ?

(A) 0.2 (B) 0.4 (C) 0.8 (D) 0.6

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Q3. MCQ (Single Correct Answer)

Match List - I with List - II

	List - I		List - II
(A)	Bronze	(I)	Cu, Ni
(B)	Brass	(II)	Fe, Cr, Ni, C
(C)	UK silver coin	(III)	Cu, Zn
(D)	Stainless steel	(IV)	Cu, Sn

Choose the correct answer from the options given below :

(A) (A)-(IV), (B)-(II), (C)-(III), (D)-(I)

(B) (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

(C) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

(D) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)

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Q4. MCQ (Single Correct Answer)

Consider a binary solution of two volatile liquid components 1 and 2. x_1 and y_1 are the mole fractions of component 1 in liquid and vapour phase, respectively. The slope and intercept of the linear plot of $\frac{1}{x_1}$ vs $\frac{1}{y_1}$ are given respectively as :

(A) $\frac{\mathrm{P}_1^0}{\mathrm{P}_2^0}, \frac{\mathrm{P}_1^0-\mathrm{P}_2^0}{\mathrm{P}_2^0}$

(B) $\frac{\mathrm{P}_2^0}{\mathrm{P}_1^0}, \frac{\mathrm{P}_2^0-\mathrm{P}_1^0}{\mathrm{P}_2^0}$

(C) $\frac{\mathrm{P}_2^0}{\mathrm{P}_1^0}, \frac{\mathrm{P}_1^0-\mathrm{P}_2^0}{\mathrm{P}_2^0}$

(D) $\frac{\mathrm{P}_1^0}{\mathrm{P}_2^0}, \frac{\mathrm{P}_2^0-\mathrm{P}_1^0}{\mathrm{P}_2^0}$

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Q5. MCQ (Single Correct Answer)

Given below are the atomic numbers of some group 14 elements. The atomic number of the element with lowest melting point is :

(A) 14 (B) 50 (C) 6 (D) 82

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Q6. MCQ (Single Correct Answer)

Given below are two statements :

Statement (I) : The boiling points of alcohols and phenols increase with increase in the number of C -atoms.

Statement (II) : The boiling points of alcohols and phenols are higher in comparison to other class of compounds such as ethers, haloalkanes.

In the light of the above statements, choose the correct answer from the options given below :

(A) Statement I is false but Statement II is true

(B) Both Statement I and Statement II are true

(C) Statement I is true but Statement II is false

(D) Both Statement I and Statement II are false

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Q7. MCQ (Single Correct Answer)

The α-Helix and β - Pleated sheet structures of protein are associated with its :

(A) tertiary structure (B) quaternary structure

(C) primary structure (D) secondary structure

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Q8. MCQ (Single Correct Answer)

The effect of temperature on spontaneity of reactions are represented as :

	ΔH	ΔS	Temperature	Spontaneity
(A)	+	−	any T	Non spontaneous
(B)	+	+	low T	spontaneous
(C)	−	−	low T	Non spontaneous
(D)	−	+	any T	spontaneous

The incorrect combinations are :

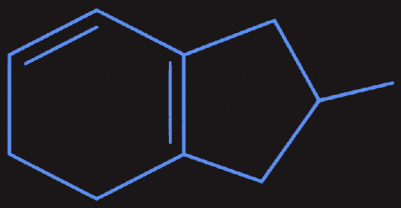
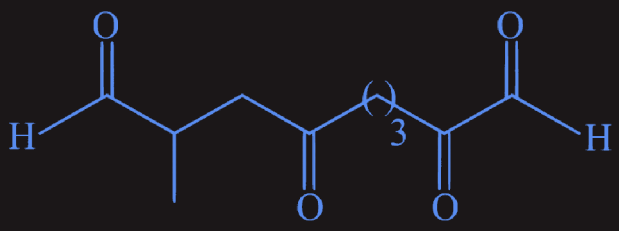
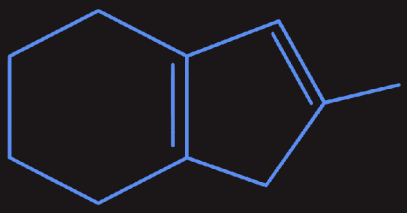
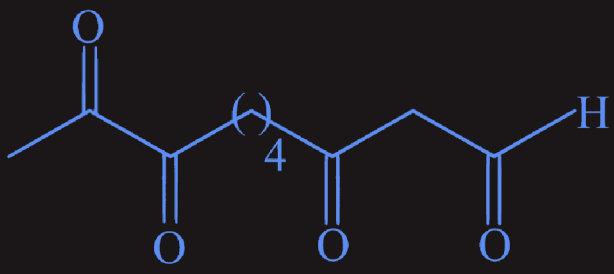
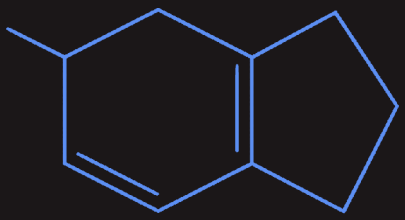
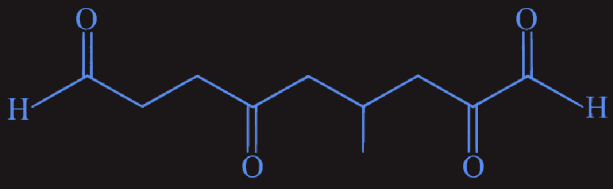
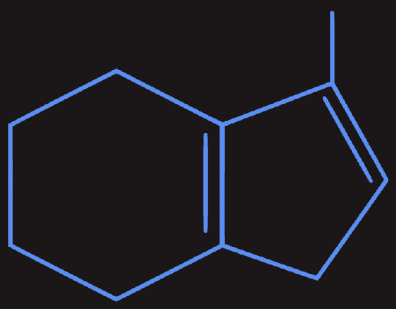
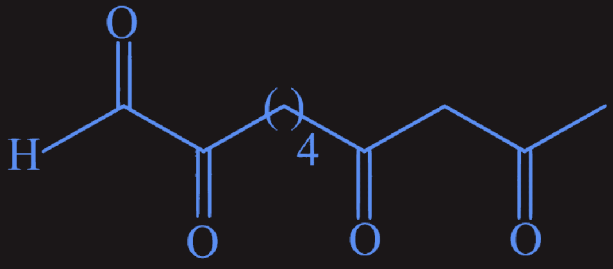
(A) (A) and (C) only (B) (B) and (D) only (C) (A) and (D) only (D) (B) and (C) only

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Q9. MCQ (Single Correct Answer)

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Match List - I with List - II

	List - I (Isomers of $C_{10}H_{14}$)		List - II (Ozonolysis product)
(A)		(I)	
(B)		(II)	
(C)		(III)	
(D)		(IV)	

Choose the correct answer from the options given below :

(A) (A) –(III), (B) – (IV), (C) – (I), (D)– (II)

(B) (A) – (I), (B) – (IV), (C) – (III), (D) – (II)

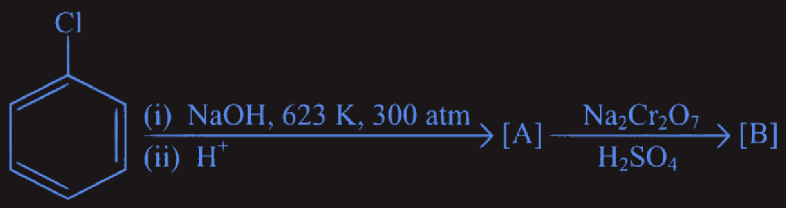
(C) (A) – (III), (B) – (II), (C) – (I), (D) – (IV)

(D) (A) – (II), (B) – (II), (C) – (I), (D) – (IV)

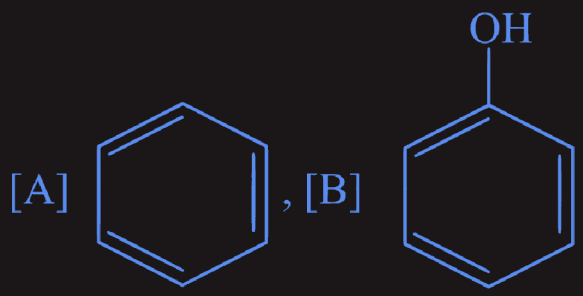
Q10. MCQ (Single Correct Answer)

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Identify the products [A] and [B], respectively in the following reaction:



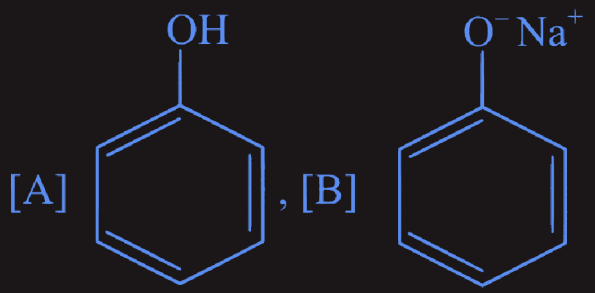
(A)



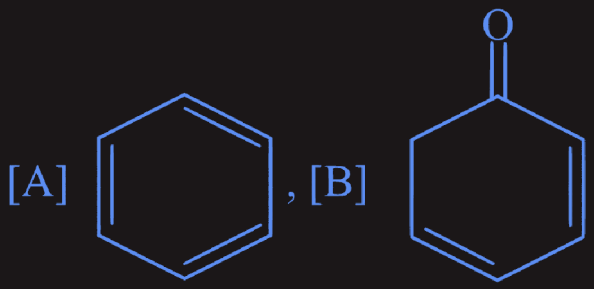
(B)



(C)



(D)

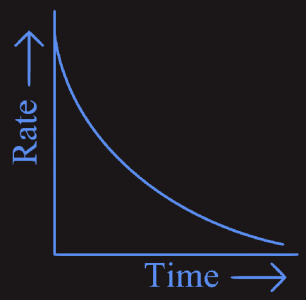


Q11. MCQ (Single Correct Answer)

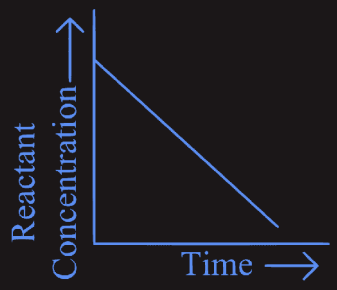
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Which of the following graphs most appropriately represents a zero order reaction?

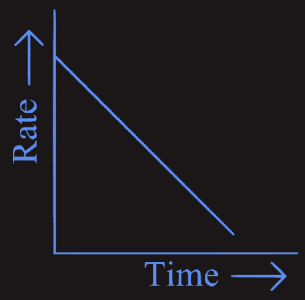
(A)



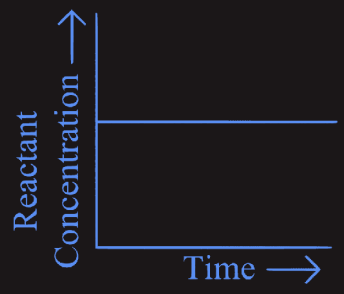
(B)



(C)



(D)



Q12. MCQ (Single Correct Answer)

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Identify the coordination complexes in which the central metal ion has d^4 configuration.

(A) $[FeO_4]^{2-}$

(B) $[Mn(CN)_6]^{3-}$

(C) $[Fe(CN)_6]^{3-}$

(D)



(E) $[NiF_6]^{2-}$

Choose the correct answer from the options given below :

(A) (B), (C) and (D) only

(B) (C) and (E) only

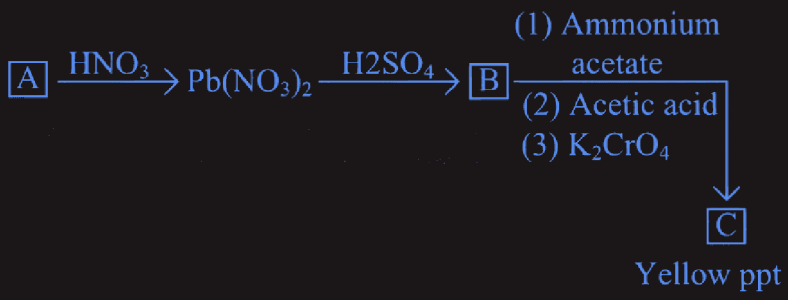
(C) $(A), (B)$ and (E) only

(D) (B) and (D) only

Q13. MCQ (Single Correct Answer)

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Identify A, B and C in the given below reaction sequence



(A) $PbCl_2, PbSO_4, PbCrO_4$

(B) $PbS, PbSO_4, Pb(CH_3COO)_2$

(C) $PbS, PbSO_4, PbCrO_4$

(D) $PbCl_2, Pb(SO_4)_2, PbCrO_4$

Q14. MCQ (Single Correct Answer)

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Standard electrode potentials for a few half cells are mentioned below :

$E^{\circ}_{Cu^{2+}/Cu} = 0.34\ V, E^{\circ}_{Zn^{2+}/Zn} = -0.76\ V$

$E^{\circ}_{Ag^{+}/Ag} = 0.80\ V, E^{\circ}_{Mg^{2+}/Mg} = -2.37\ V$

Which one of the following cells gives the most negative value of ΔG° ?

(A) $Zn\left|Zn^{2+}(1M)\right|\left|Ag^{+}(1M)\right|Ag$

(B) $Ag\left|Ag^{+}(1M)\right|\left|Mg^{2+}(1M)\right|Mg$

(C) $Zn\left|Zn^{2+}(1M)\right|\left|Mg^{2+}(1M)\right|Mg$

(D) $Cu\left|Cu^{2+}(1M)\right|\left|Ag^{+}(1M)\right|Ag$

Q15. MCQ (Single Correct Answer)

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Consider the reaction

$X_2Y(g) \rightleftharpoons X_2(g) + \frac{1}{2}Y_2(g)$

The equation representing correct relationship between the degree of dissociation (x) of $X_2Y(g)$ with its equilibrium constant Kp is _____.

Assume x to be very very small.

(A) $x = \sqrt[3]{\frac{Kp}{p}}$

(B) $x = \sqrt[3]{\frac{Kp}{2p}}$

(C) $x = \sqrt[3]{\frac{2Kp^2}{p}}$

(D) $x = \sqrt[3]{\frac{2Kp}{p}}$

Q16. MCQ (Single Correct Answer)

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Given below are two statements :

Statement (I) : For a given shell, the total number of allowed orbitals is given by n^2.

Statement (II) : For any subshell, the spatial orientation of the orbitals is given by $-l$ to $+l$ values including zero.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

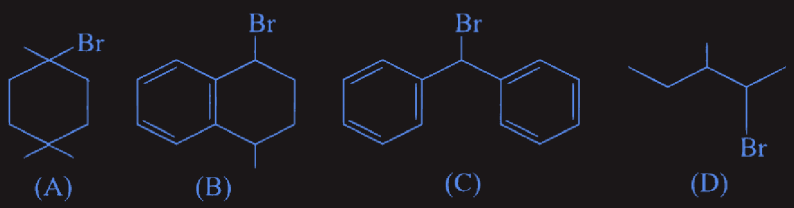
(C) Statement I is false but Statement II is true

(D) Statement I is true but Statement II is false

Q17. MCQ (Single Correct Answer)

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The ascending order of relative rate of solvolysis of following compounds is :



(A) $(C) < (D) < (B) < (A)$

(B) $(D) < (A) < (B) < (C)$

(C) $(D) < (B) < (A) < (C)$

(D) $(C) < (B) < (A) < (D)$

Q18. MCQ (Single Correct Answer)

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Given below are two statements about X-ray spectra of elements :

Statement (I) : A plot of $\sqrt{\nu}$ ($\nu =$ frequency of X-rays emitted) vs atomic mass is a straight line.

Statement (II) : A plot of ν ($\nu =$ frequency of X-rays emitted) vs atomic number is a straight line. In the light of the above statements, choose the correct answer from the options given below :

(A) Statement I is false but Statement II is true

(B) Both Statement I and Statement II are true

(C) Statement I is true but Statement II is false

(D) Both Statement I and Statement II are false

Q19. MCQ (Single Correct Answer)

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Consider the following reactions

$$K_2Cr_2O_7 \xrightarrow[-H_2O]{KOH} [A] \xrightarrow[-H_2O]{H_2SO_4} [B] + K_2SO_4$$

The products $[A]$ and $[B]$, respectively are :

(A) $K_2Cr(OH)_6$ and Cr_2O_3

(B) K_2CrO_4 and Cr_2O_3

(C) K_2CrO_4 and $K_2Cr_2O_7$

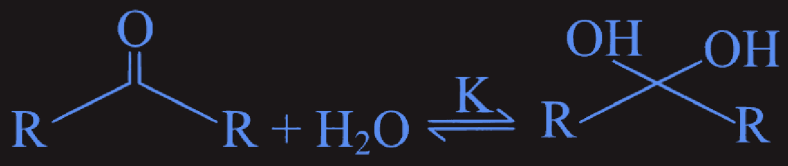
(D) K_2CrO_4 and CrO

Q20. MCQ (Single Correct Answer)

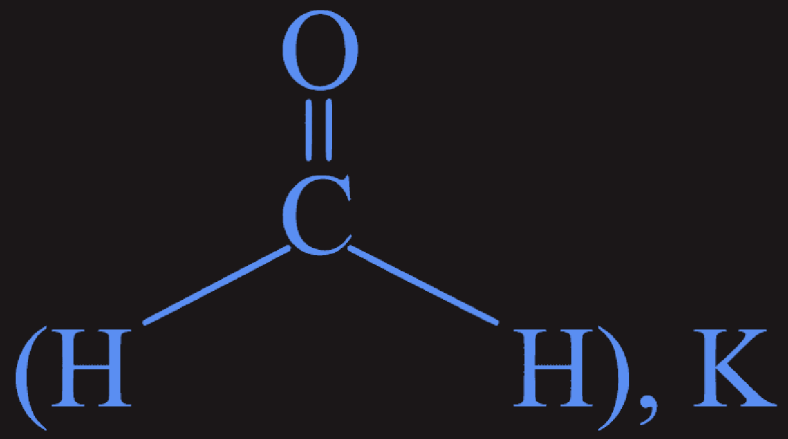
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Given below are two statements:

Consider the following reaction

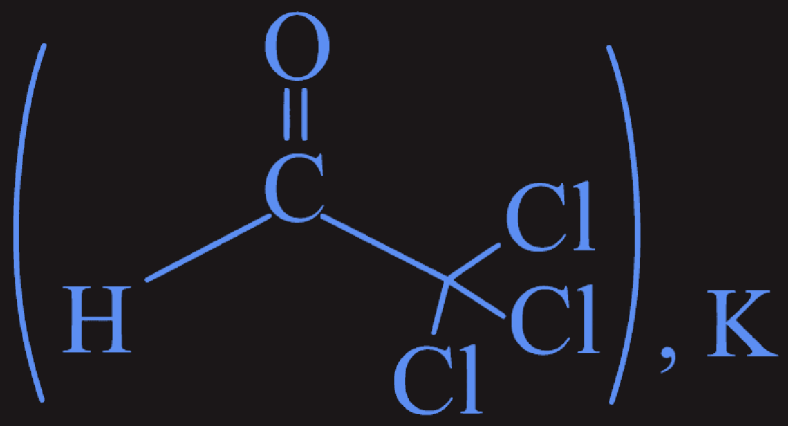


Statement (I): In the case of formaldehyde



is about 2280, due to small substituents, hydration is faster.

Statement (II) : In the case of trichloro acetaldehyde



is about 2000 due to $-I$ effect of $-Cl$.

In the light of the above statements, choose the correct answer from the options given below :

(A) Statement I is true but Statement II is false

(B) Both Statement I and Statement II are true

(C) Statement I is false but Statement II is true

(D) Both Statement I and Statement II are false

Q21. Numerical

0.01 mole of an organic compound (X) containing 10% hydrogen, on complete combustion produced $0.9\ \mathrm{g}\ \mathrm{H_2O}$. Molar mass of (X) is _____ $\mathrm{g\ mol^{-1}}$.

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Q22. Numerical

The bond dissociation enthalpy of $\mathrm{X_2}\Delta\mathrm{H}^\circ_{\text{bond}}$ calculated from the given data is _____ $\mathrm{kJmol^{-1}}$. (Nearest integer)

$\mathrm{M^+X^-(s) \rightarrow M^+(g) + X^-(g)}\Delta\mathrm{H}^\circ_{\text{lattice}} = 800\ \mathrm{kJ\ mol^{-1}}$

$\mathrm{M(\ s) \rightarrow M(\ g)}\Delta\mathrm{H}^\circ_{\text{sub}} = 100\ \mathrm{kJ\ mol^{-1}}$

$\mathrm{M(\ g) \rightarrow M^+(g) + e^-(g)}\Delta\mathrm{H}^\circ_{\text{i}} = 500\ \mathrm{kJ\ mol^{-1}}$

$\mathrm{X(\ g) + e^-(g) \rightarrow X^-(g)}\Delta\mathrm{H}^\circ_{\text{eg}} = -300\ \mathrm{kJ\ mol^{-1}}$

$\mathrm{M(\ s) + \frac{1}{2}X_2(\ g) \rightarrow M^+X^-(s)}\Delta\mathrm{H}^\circ_f = -400\ \mathrm{kJ\ mol^{-1}}$

[Given : $\mathrm{M^+X^-}$ is a pure ionic compound and X forms a diatomic molecule $\mathrm{X_2}$ in gaseous state]

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Q23. Numerical

When 81.0 g of aluminium is allowed to react with 128.0 g of oxygen gas, the mass of aluminium oxide produced in grams is _____ . (Nearest integer)

Given :

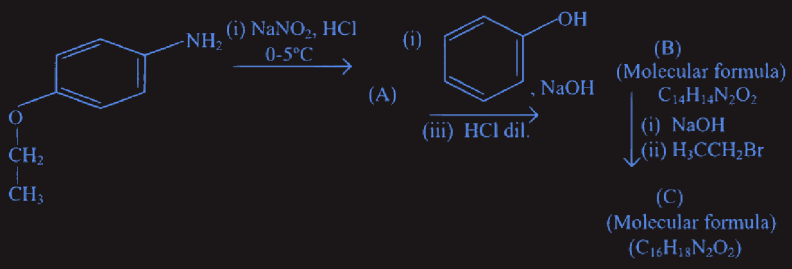
Molar mass of Al is $27.0\ \mathrm{g\ mol^{-1}}$

Molar mass of O is $16.0\ \mathrm{g\ mol^{-1}}$

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Q24. Numerical

Consider the following sequence of reactions.



Total number of $\mathrm{sp^3}$ hybridised carbon atoms in the major product C formed is _____.

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Q25. Numerical

A compound ' X ' absorbs 2 moles of hydrogen and ' X ' upon oxidation with $\mathrm{KMnO_4}\ |\ \mathrm{H^+}$ gives

$\mathrm{CH_3-\overset{\overset{O}{\|}}{C}-CH_3}$, $\mathrm{CH_3-\overset{\overset{O}{\|}}{C}-OH}$ and $\mathrm{CH_3-\overset{\overset{O}{\|}}{C}-CH_2CH_2-\overset{\overset{O}{\|}}{C}-OH}$.

The total number of σ bonds present in the compound ' X ' is _____.

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Mathematics 25 questions

Q26. MCQ (Single Correct Answer)

If $I=\int_0^{\frac{\pi}{2}} \frac{\sin^{\frac{3}{2}} x}{\sin^{\frac{3}{2}} x+\cos^{\frac{3}{2}} x} dx$, then $\int_0^{2I} \frac{x \sin x \cos x}{\sin^4 x+\cos^4 x} dx$ equals :

(A) $\frac{\pi^2}{12}$ (B) $\frac{\pi^2}{4}$ (C) $\frac{\pi^2}{16}$ (D) $\frac{\pi^2}{8}$

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Q27. MCQ (Single Correct Answer)

The number of complex numbers z, satisfying $|z|=1$ and $\left|\frac{z}{\bar{z}}+\frac{\bar{z}}{z}\right|=1$, is :

(A) 8 (B) 10 (C) 4 (D) 6

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Q28. MCQ (Single Correct Answer)

The length of the chord of the ellipse $\frac{x^2}{4}+\frac{y^2}{2}=1$, whose mid-point is $\left(1, \frac{1}{2}\right)$, is :

(A) $\frac{2}{3}\sqrt{15}$ (B) $\frac{1}{3}\sqrt{15}$ (C) $\sqrt{15}$ (D) $\frac{5}{3}\sqrt{15}$

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Q29. MCQ (Single Correct Answer)

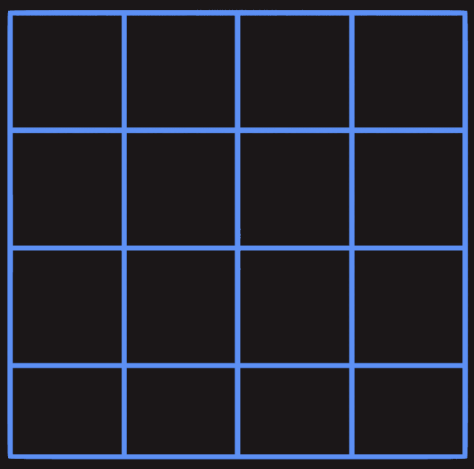
Let the point A divide the line segment joining the points $P(-1,-1,2)$ and $Q(5,5,10)$ internally in the ratio $r:1(r>0)$. If O is the origin and $(\overrightarrow{OQ}\cdot\overrightarrow{OA})-\frac{1}{5}|\overrightarrow{OP}\times\overrightarrow{OA}|^2=10$, then the value of r is :

(A) $\sqrt{7}$ (B) 14 (C) 7 (D) 3

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Q30. MCQ (Single Correct Answer)

A board has 16 squares as shown in the figure :



Out of these 16 squares, two squares are chosen at random. The probability that they have no side in common is :

(A) $\frac{3}{5}$ (B) $\frac{4}{5}$ (C) $\frac{23}{30}$ (D) $\frac{7}{10}$

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Q31. MCQ (Single Correct Answer)

If the square of the shortest distance between the lines $\frac{x-2}{1}=\frac{y-1}{2}=\frac{z+3}{-3}$ and $\frac{x+1}{2}=\frac{y+3}{4}=\frac{z+5}{-5}$ is $\frac{m}{n}$, where m, n are coprime numbers, then $m+n$ is equal to :

(A) 14 (B) 6 (C) 21 (D) 9

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Q32. MCQ (Single Correct Answer)

Let $\mathrm{X}=\mathbf{R}\times\mathbf{R}$. Define a relation R on X as :

$(a_1,b_1)R(a_2,b_2)\Leftrightarrow b_1=b_2$

Statement I: R is an equivalence relation.

Statement II : For some $(\mathrm{a},\mathrm{b})\in\mathrm{X}$, the set $\mathrm{S}=\{(x,y)\in\mathrm{X}:(x,y)\mathrm{R}(\mathrm{a},\mathrm{b})\}$ represents a line parallel to $y=x$.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are true

(B) Statement I is true but Statement II is false

(C) Both Statement I and Statement II are false

(D) Statement I is false but Statement II is true

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Q33. MCQ (Single Correct Answer)

If the area of the region $\left\{(x,y):-1\le x\le 1, 0\le y\le \mathrm{a}+\mathrm{e}^{|x|}-\mathrm{e}^{-x}, \mathrm{a}>0\right\}$ is $\frac{\mathrm{e}^2+8\mathrm{e}+1}{\mathrm{e}}$, then the value of a is :

(A) 7 (B) 5 (C) 6 (D) 8

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Q34. MCQ (Single Correct Answer)

Let the range of the function $f(x)=6+16\cos x\cdot\cos\left(\frac{\pi}{3}-x\right)\cdot\cos\left(\frac{\pi}{3}+x\right)\cdot\sin 3x\cdot\cos 6x, x\in\mathbf{R}$ be $[\alpha,\beta]$. Then the distance of the point (α,β) from the line $3x+4y+12=0$ is :

(A) 11 (B) 10 (C) 8 (D) 9

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Q35. MCQ (Single Correct Answer)

The system of equations

$x+y+z=6,$
$x+2y+5z=9,$
$x+5y+\lambda z=\mu,$

has no solution if

(A) $\lambda=17,\mu=18$ (B) $\lambda=17,\mu\neq 18$ (C) $\lambda=15,\mu\neq 17$ (D) $\lambda\neq 17,\mu\neq 18$

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Q36. MCQ (Single Correct Answer)

The distance of the line $\frac{x-2}{2}=\frac{y-6}{3}=\frac{z-3}{4}$ from the point $(1,4,0)$ along the line $\frac{x}{1}=\frac{y-2}{2}=\frac{z+3}{3}$ is :

(A) $\sqrt{17}$ (B) $\sqrt{13}$ (C) $\sqrt{15}$ (D) $\sqrt{14}$

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Q37. MCQ (Single Correct Answer)

$\lim\limits_{x\to\infty}\frac{(2x^2-3x+5)(3x-1)^{\frac{x}{2}}}{(3x^2+5x+4)\sqrt{(3x+2)^x}}$ is equal to :

(A) $\frac{2e}{3}$ (B) $\frac{2}{3\sqrt{e}}$ (C) $\frac{2e}{\sqrt{3}}$ (D) $\frac{2}{\sqrt{3e}}$

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Q38. MCQ (Single Correct Answer)

Let $A=[a_{ij}]$ be a 3×3 matrix such that $A\begin{bmatrix}0\\1\\0\end{bmatrix}=\begin{bmatrix}0\\0\\1\end{bmatrix}, A\begin{bmatrix}4\\1\\3\end{bmatrix}=\begin{bmatrix}0\\1\\0\end{bmatrix}$ and $A\begin{bmatrix}2\\1\\2\end{bmatrix}=\begin{bmatrix}1\\0\\0\end{bmatrix}$, then a_{23} equals :

(A) 2 (B) -1 (C) 1 (D) 0

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Q39. MCQ (Single Correct Answer)

Let $\mathrm{A}=\{(x,y)\in\mathbf{R}\times\mathbf{R}:|x+y|\geqslant 3\}$ and $\mathrm{B}=\{(x,y)\in\mathbf{R}\times\mathbf{R}:|x|+|y|\leq 3\}$. If $\mathrm{C}=\{(x,y)\in\mathrm{A}\cap\mathrm{B}:x=0 \text{ or } y=0\}$, then $\sum_{(x,y)\in\mathrm{C}}|x+y|$ is :

(A) 18 (B) 24 (C) 15 (D) 12

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Q40. MCQ (Single Correct Answer)

If in the expansion of $(1+x)^{\mathrm{p}}(1-x)^{\mathrm{q}}$, the coefficients of x and x^2 are 1 and -2 , respectively, then $\mathrm{p}^2+\mathrm{q}^2$ is equal to :

(A) 8 (B) 20 (C) 13 (D) 18

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Q41. MCQ (Single Correct Answer)

A rod of length eight units moves such that its ends A and B always lie on the lines $x-y+2=0$ and $y+2=0$, respectively. If the locus of the point P, that divides the rod AB internally in the ratio $2:1$ is $9\left(x^2+\alpha y^2+\beta xy+\gamma x+28y\right)-76=0$, then $\alpha-\beta-\gamma$ is equal to :

(A) 24 (B) 22 (C) 21 (D) 23

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Q42. MCQ (Single Correct Answer)

Let $\int x^3\sin x\,dx=g(x)+C$, where C is the constant of integration. If $8\left(g\left(\frac{\pi}{2}\right)+g'\left(\frac{\pi}{2}\right)\right)=\alpha\pi^3+\beta\pi^2+\gamma, \alpha,\beta,\gamma\in Z$, then $\alpha+\beta-\gamma$ equals :

(A) 47 (B) 55 (C) 62 (D) 48

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Q43. MCQ (Single Correct Answer)

A spherical chocolate ball has a layer of ice-cream of uniform thickness around it. When the thickness of the ice-cream layer is 1 cm , the ice-cream melts at the rate of $81\ \mathrm{cm}^3/\mathrm{min}$ and the thickness of the ice-cream layer decreases at the rate of $\frac{1}{4\pi}\ \mathrm{cm}/\mathrm{min}$. The surface area (in cm^2) of the chocolate ball (without the ice-cream layer) is :

(A) 128π (B) 196π (C) 225π (D) 256π

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Q44. MCQ (Single Correct Answer)

Let the shortest distance from $(a, 0), a > 0$, to the parabola $y^2 = 4x$ be 4 . Then the equation of the circle passing through the point $(a, 0)$ and the focus of the parabola, and having its centre on the axis of the parabola is :

(A) $x^2 + y^2 - 8x + 7 = 0$ (B) $x^2 + y^2 - 6x + 5 = 0$

(C) $x^2 + y^2 - 4x + 3 = 0$ (D) $x^2 + y^2 - 10x + 9 = 0$

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Q45. MCQ (Single Correct Answer)

Let $x = x(y)$ be the solution of the differential equation $y = \left(x - y\frac{\mathrm{d}x}{\mathrm{d}y}\right)\sin\left(\frac{x}{y}\right), y > 0$ and $x(1) = \frac{\pi}{2}$. Then $\cos(x(2))$ is equal to :

(A) $2\left(\log_e 2\right) - 1$ (B) $1 - 2(\log_e 2)^2$

(C) $1 - 2\left(\log_e 2\right)$ (D) $2(\log_e 2)^2 - 1$

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Q46. Numerical

Let α, β be the roots of the equation $x^2 - \mathrm{ax} - \mathrm{b} = 0$ with $\mathrm{Im}(\alpha) < \mathrm{Im}(\beta)$. Let $\mathrm{P_n} = \alpha^{\mathrm{n}} - \beta^{\mathrm{n}}$. If $\mathrm{P_3} = -5\sqrt{7}i, \mathrm{P_4} = -3\sqrt{7}i, \mathrm{P_5} = 11\sqrt{7}i$ and $\mathrm{P_6} = 45\sqrt{7}i$, then $\left|\alpha^4 + \beta^4\right|$ is equal to _____.

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Q47. Numerical

The focus of the parabola $y^2 = 4x + 16$ is the centre of the circle C of radius 5 . If the values of λ, for which C passes through the point of intersection of the lines $3x - y = 0$ and $x + \lambda y = 4$, are λ_1 and $\lambda_2, \lambda_1 < \lambda_2$, then $12\lambda_1 + 29\lambda_2$ is equal to _____ .

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Q48. Numerical

The number of ways, 5 boys and 4 girls can sit in a row so that either all the boys sit together or no two boys sit together, is _____.

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Q49. Numerical

The roots of the quadratic equation $3x^2 - px + q = 0$ are 10th and 11th terms of an arithmetic progression with common difference $\frac{3}{2}$. If the sum of the first 11 terms of this arithmetic progression is 88 , then $q - 2p$ is equal to _____ .

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Q50. Numerical

The variance of the numbers $8, 21, 34, 47, \ldots, 320$ is _____.

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Physics 25 questions

Q51. MCQ (Single Correct Answer)

The equation of a transverse wave travelling along a string is $y(x,t) = 4.0 \sin\left[20 \times 10^{-3}x + 600t\right]\text{mm}$, where x is in mm and t is in second. The velocity of the wave is :

(A) $-60\ \text{m/s}$ (B) $+60\ \text{m/s}$ (C) $+30\ \text{m/s}$ (D) $-30\ \text{m/s}$

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Q52. MCQ (Single Correct Answer)

A plane electromagnetic wave of frequency 20 MHz travels in free space along the $+x$ direction. At a particular point in space and time, the electric field vector of the wave is $\text{E}_y = 9.3\text{Vm}^{-1}$. Then, the magnetic field vector of the wave at that point is

(A) $\text{B}_z = 1.55 \times 10^{-8}\ \text{T}$ (B) $\text{B}_z = 6.2 \times 10^{-8}\ \text{T}$

(C) $\text{B}_z = 3.1 \times 10^{-8}\ \text{T}$ (D) $\text{B}_z = 9.3 \times 10^{-8}\ \text{T}$

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Q53. MCQ (Single Correct Answer)

If a satellite orbiting the Earth is 9 times closer to the Earth than the Moon, what is the time period of rotation of the satellite? Given rotational time period of Moon $= 27$ days and gravitational attraction between the satellite and the moon is neglected.

(A) 3 days (B) 27 days (C) 81 days (D) 1 day

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Q54. MCQ (Single Correct Answer)

A massless spring gets elongated by amount x_1 under a tension of 5 N . Its elongation is x_2 under the tension of 7 N . For the elongation of $(5x_1 - 2x_2)$, the tension in the spring will be,

(A) 20 N (B) 39 N (C) 11 N (D) 15 N

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Q55. MCQ (Single Correct Answer)

Water of mass m gram is slowly heated to increase the temperature from T_1 to T_2. The change in entropy of the water, given specific heat of water is $1\mathrm{Jkg}^{-1}\,\mathrm{K}^{-1}$, is :

(A) $\mathrm{m}\,(\mathrm{T}_2-\mathrm{T}_1)$

(B) zero

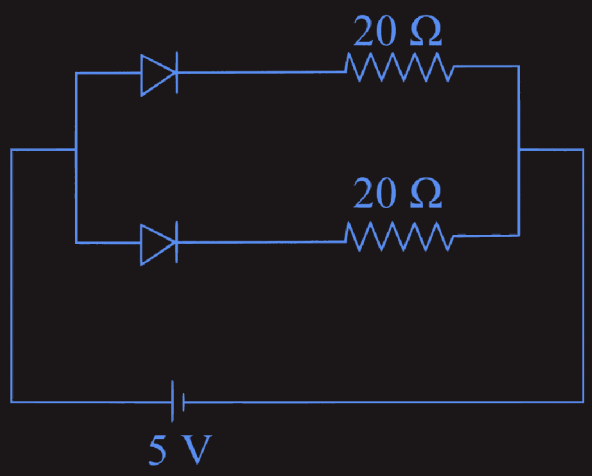
(C) $\mathrm{m}\ln\left(\frac{\mathrm{T}_1}{\mathrm{T}_2}\right)$

(D) $\mathrm{m}\ln\left(\frac{\mathrm{T}_2}{\mathrm{T}_1}\right)$

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Q56. MCQ (Single Correct Answer)

What is the current through the battery in the circuit shown below?



(A) 1.5 A

(B) 0.25 A

(C) 1.0 A

(D) 0.5 A

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Q57. MCQ (Single Correct Answer)

Given below are two statements. One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : The binding energy per nucleon is found to be practically independent of the atomic number A , for nuclei with mass numbers between 30 and 170 .

Reason (R) : Nuclear force is long range. In the light of the above statements, choose the correct answer from the options given below :

(A) (A) is true but (R) is false

(B) Both (A) and (R) are true and (R) is the correct explanation of (A)

(C) (A) is false but (R) is true

(D) Both (A) and (R) are true but (R) is NOT the correct explanation of (A)

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Q58. MCQ (Single Correct Answer)

A ball having kinetic energy KE, is projected at an angle of 60° from the horizontal. What will be the kinetic energy of ball at the highest point of its flight?

(A) $\frac{(\mathrm{KE})}{2}$

(B) $\frac{(\mathrm{KE})}{8}$

(C) $\frac{(\mathrm{KE})}{4}$

(D) $\frac{(\mathrm{KE})}{16}$

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Q59. MCQ (Single Correct Answer)

In photoelectric effect an em-wave is incident on a metal surface and electrons are ejected from the surface. If the work function of the metal is 2.14 eV and stopping potential is 2 V , what is the wavelength of the em-wave? (Given $\mathrm{hc} = 1242\mathrm{eVnm}$ where h is the Planck's constant and c is the speed of light in vacuum.)

(A) 400 nm (B) 600 nm (C) 300 nm (D) 200 nm

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Q60. MCQ (Single Correct Answer)

Two point charges $-4\mu c$ and $4\mu c$, constituting an electric dipole, are placed at $(-9, 0, 0)\mathrm{cm}$ and $(9, 0, 0)\mathrm{cm}$ in a uniform electric field of strength $10^4\mathrm{NC}^{-1}$. The work done on the dipole in rotating it from the equilibrium through 180° is :

(A) 18.4 mJ (B) 12.4 mJ (C) 16.4 mJ (D) 14.4 mJ

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Q61. MCQ (Single Correct Answer)

A concave mirror of focal length f in air is dipped in a liquid of refractive index μ. Its focal length in the liquid will be:

(A) μf (B) f (C) $\frac{f}{\mu}$ (D) $\frac{f}{(\mu-1)}$

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Q62. MCQ (Single Correct Answer)

The width of one of the two slits in Young's double slit experiment is d while that of the other slit is x d. If the ratio of the maximum to the minimum intensity in the interference pattern on the screen is $9:4$ then what is the value of x ? (Assume that the field strength varies according to the slit width.)

(A) 2 (B) 3 (C) 4 (D) 5

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Q63. MCQ (Single Correct Answer)

The refractive index of the material of a glass prism is $\sqrt{3}$. The angle of minimum deviation is equal to the angle of the prism. What is the angle of the prism?

(A) 60° (B) 50° (C) 58° (D) 48°

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Q64. MCQ (Single Correct Answer)

Two charges $7\mu c$ and $-4\mu c$ are placed at $(-7\ \mathrm{cm}, 0, 0)$ and $(7\ \mathrm{cm}, 0, 0)$ respectively. Given, $\epsilon_0 = 8.85 \times 10^{-12}\mathrm{C}^2\ \mathrm{N}^{-1}\ \mathrm{m}^{-2}$, the electrostatic potential energy of the charge configuration is :

(A) -2.0 J (B) -1.5 J (C) -1.2 J (D) -1.8 J

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Q65. MCQ (Single Correct Answer)

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Water flows in a horizontal pipe whose one end is closed with a valve. The reading of the pressure gauge attached to the pipe is P_1. The reading of the pressure gauge falls to P_2 when the valve is opened. The speed of water flowing in the pipe is proportional to

(A) $(P_1 - P_2)^2$ (B) $\sqrt{P_1 - P_2}$ (C) $P_1 - P_2$ (D) $(P_1 - P_2)^4$

Q66. MCQ (Single Correct Answer)

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The energy of a system is given as $E(t) = \alpha^3 e^{-\beta t}$, where t is the time and $\beta = 0.3\,s^{-1}$. The errors in the measurement of α and t are 1.2% and 1.6%, respectively. At $t = 5\,s$, maximum percentage error in the energy is :

(A) 6% (B) 11.6% (C) 4% (D) 8.4%

Q67. MCQ (Single Correct Answer)

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Match List - I with List - II

	List - I		List - II
(A)	Permeability of free space	(I)	$[ML^2T^{-2}]$
(B)	Magnetic field	(II)	$[MT^{-2}A^{-1}]$
(C)	Magnetic moment	(III)	$[MLT^{-2}A^{-2}]$
(D)	Torsional constant	(IV)	$[L^2A]$

Choose the correct answer from the options given below :

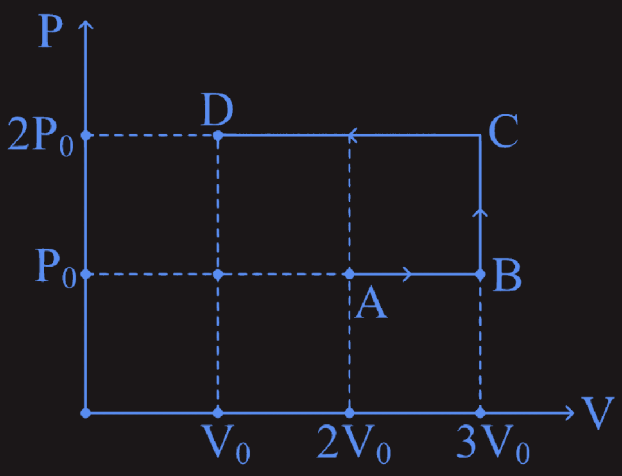
(A) (A)-(III), (B)-(II), (C)-(IV), (D)-(I)

(B) $(A) - (I), (B) - (IV), (C) - (II), (D) - (III)$

(C) $(A) - (II), (B) - (I), (C) - (III), (D) - (IV)$

(D) (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

Q68. MCQ (Single Correct Answer)



Using the given $P-V$ diagram, the work done by an ideal gas along the path $ABCD$ is :

(A) $3\mathrm{P}_0\,\mathrm{V}_0$ (B) $-3\mathrm{P}_0\,\mathrm{V}_0$ (C) $-4\mathrm{P}_0\,\mathrm{V}_0$ (D) $4\mathrm{P}_0\,\mathrm{V}_0$

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Q69. MCQ (Single Correct Answer)

A circular disk of radius R meter and mass M kg is rotating around the axis perpendicular to the disk. An external torque is applied to the disk such that $\theta(t)=5t^2-8t$, where $\theta(t)$ is the angular position of the rotating disc as a function of time t. How much power is delivered by the applied torque, when $t=2\,\mathrm{s}$?

(A) $60\mathrm{MR}^2$ (B) $72\mathrm{MR}^2$ (C) $8\mathrm{MR}^2$ (D) $108\mathrm{MR}^2$

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Q70. MCQ (Single Correct Answer)

A galvanometer having a coil of resistance 30Ω need 20 mA of current for full-scale deflection. If a maximum current of 3 A is to be measured using this galvanometer, the resistance of the shunt to be added to the galvanometer should be $\frac{30}{X}\Omega$, where X is

(A) 447 (B) 298 (C) 149 (D) 596

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Q71. Numerical

A time varying potential difference is applied between the plates of a parallel plate capacitor of capacitance $2.5\mu\,\mathrm{F}$. The dielectric constant of the medium between the capacitor plates is 1 . It produces an instantaneous displacement current of 0.25 mA in the intervening space between the capacitor plates, the magnitude of the rate of change of the potential difference will be _____ Vs^{-1}.

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Q72. Numerical

In a series LCR circuit, a resistor of 300Ω, a capacitor of 25 nF and an inductor of 100 mH are used. For maximum current in the circuit, the angular frequency of the ac source is _____ $\times 10^4$ radians s^{-1}

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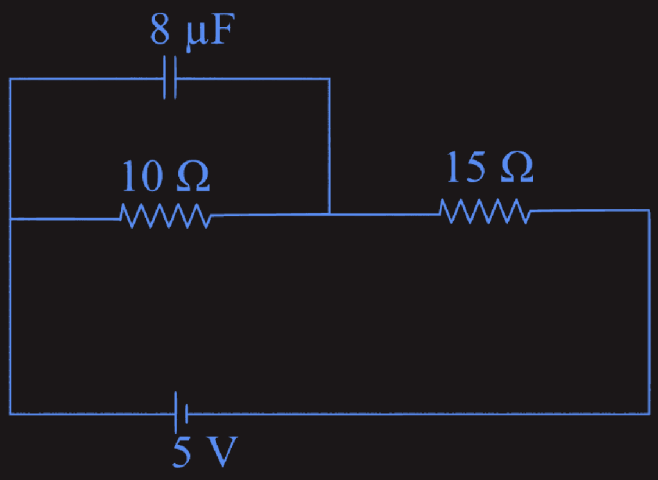
Q73. Numerical

An air bubble of radius 1.0 mm is observed at a depth 20 cm below the free surface of a liquid having surface tension $0.095\ \mathrm{J/m^2}$ and density $10^3\ \mathrm{kg/m^3}$. The difference between pressure inside the bubble and atmospheric pressure is _____ $\mathrm{N/m^2}$. (Take $\mathrm{g} = 10\ \mathrm{m/s^2}$)

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Q74. Numerical

At steady state the charge on the capacitor, as shown in the circuit below, is _____ μC.



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Q75. Numerical

A satellite of mass $\frac{M}{2}$ is revolving around earth in a circular orbit at a height of $\frac{R}{3}$ from earth surface. The angular momentum of the satellite is $\mathrm{M}\sqrt{\frac{\mathrm{GMR}}{x}}$. The value of x is _____ , where M and R are the mass and radius of earth, respectively. (G is the gravitational constant)

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Answer Key

Chemistry

1. **A**	2. **D**	3. **B**	4. **D**	5. **B**
6. **B**	7. **D**	8. **D**	9. **A**	10. **B**
11. **B**	12. **D**	13. **C**	14. **A**	15. **C**
16. **A**	17. **B**	18. **D**	19. **C**	20. **B**
21. **100**	22. **200**	23. **153**	24. **4**	25. **27**

Mathematics

26. **C**	27. **A**	28. **A**	29. **C**	30. **B**
31. **D**	32. **B**	33. **B**	34. **A**	35. **B**
36. **D**	37. **B**	38. **B**	39. **D**	40. **C**
41. **D**	42. **B**	43. **D**	44. **B**	45. **D**
46. **31**	47. **15**	48. **17280**	49. **474**	50. **8788**

Physics

51. **D**	52. **C**	53. **D**	54. **C**	55. **D**
56. **D**	57. **A**	58. **C**	59. **C**	60. **D**
61. **B**	62. **D**	63. **A**	64. **D**	65. **B**
66. **A**	67. **A**	68. **B**	69. **A**	70. **C**
71. **100**	72. **2**	73. **2190**	74. **16**	75. **3**

For numerical answers given as a range, any value within the range is correct.

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